

Integrating UNCTAD T&T dataset into FlowMax

Technical report

1. Background

Unit logistics costs (\$/ton.km) are one of the key inputs for FlowMax to assess the impact of transport policy measures. Using this input, FlowMax can be utilized to compute shipper's logistics costs over an international route which represent the total monetary cost incurred by shippers for transporting goods from the origin to the destination by commodity and mode of transport. This encompasses the entire multimodal transport chain, which for international maritime transport includes hinterland transport between the origin city and the port of origin, as well as between the destination city and the port of destination. The components of logistics costs are outlined as follows:

- Maritime transport and time costs between port of origin and destination
- Hinterland transport and time costs from an origin and to a destination
- Handling costs due to transshipment and transloading operation in ports or terminals
- The interest costs of the capital goods during transport
- Potential depreciation of the value of the goods during transit
- Potential loss of opportunities due to shortage of supply for the market
- Disruption in the production process due to delayed arrival of raw materials
- Opportunities to minimize inventory costs by adjusting their distribution structure

The calculation of logistics costs serves multiple purposes. For instance, they can be used to measure the total savings in logistics costs resulting from shorter travel distances due to infrastructure upgrades. Additionally, logistics costs provide valuable insights into a country's competitiveness in the global trading market. Furthermore, changes in logistics costs, caused by policy scenarios, can be used to assess the impact on trade volumes and the broader economic environment. This is due to the strong correlation between logistics costs and trade activity.

The methodology currently employed in FlowMax to calculate logistics costs is based on several key inputs—unit transport cost per distance (\$/tonne-km), and value of time. The multiplication of unit transport cost per distance and transport distance results in

transport costs per tonne. Time costs is calculated as the product of the Value of Time (\$/tonne.hour) and transport time(hour). The addition between transport costs per tonne and time costs per tonne represents the unit logistics cost per tonne.

Currently, unit transport costs per distance and Value of Time are differentiated only by commodity type. However, this represents a simplified assumption, as unit logistics costs per distance are influenced by many factors such as export-import country pairs and the mode of transport, the quality of port and infrastructure of the countries, the efficiency of logistics and custom processes. For example, countries with more efficient infrastructure are expected to exhibit lower unit costs per distance, while air transport typically incurs the highest unit costs compared to other modes such as rail, road, and sea.

The launch of UNCTAD T&T dataset in May 2024¹ has provided an opportunity to update FlowMax' database on transport costs and to enable a more robust analysis of logistics costs of imports and exports by the model. To date, the UNCTAD T&T dataset is the only dataset which breaks down transport costs based on 4 major modes and covers extended number of countries worldwide (i.e. 222 countries as exporters and 169 as importers) for the year 2016 to 2021. It also disaggregates transport costs by origin, destination, and commodities on HS-2 digit level.

However, despite the extensive coverage of UNCTAD T&T dataset, there are still substantial transport costs data which are missing globally. Overall, 72% of the full dataset is missing due to absence of data for specific country pairs, commodity, and mode. Furthermore, the UNCTAD T&T records the transport costs in terms of CIF-FOB margin (in \$) which does not take into account the transport costs and time costs of hinterland transport such as port of arrival to the destination. Therefore, to amend this gap, developed an imputation method to estimate the missing data and include the port-hinterland costs,

¹ Launch information can be found on: <https://unctad.org/news/new-global-dataset-reveals-hidden-costs-international-trade-and-transport>

This report presents an updated methodology for imputing the missing data in UNCTAD T&T dataset and to integrate these imputed data on unit logistics costs into FlowMax. The objective of this project is to capture greater variability in unit costs per distance across exporting and importing countries, as well as different transportation modes, to enhance FlowMax' robustness in assessing impacts of transport policy scenarios.

The rest of this report is as follows, we present a review on different trade and logistics costs databases to justify the selection of UNTACD T&T dataset as the primary data source. Next, the methodology for integrating the UNTACD T&T dataset into FlowMax is outlined, detailing how it enhances the calculation of unit logistics costs. Finally, the key data employed in the method and their respective sources are introduced.

2. Review on trade and logistics costs databases

In this section we present an overview of relevant databases on logistics costs and trade published by various organizations. The summary of the characteristics of these datasets along with the modelling used to impute any missing data is presented in Table 1.

Table 1 Overview of transport and trade databases

	ECLAC database CEPALSTAT	OECD maritime transport database	OECD International Transport and Insurance Costs (ITIC) database	UNCTAD Trade and Logistics costs dataset	WB-UNESCAP trade costs database
Data sources	Import and export customs of the Latin American Integration Association (ALADI). Data sources are primarily from 10 ALADI countries and UN COMTRADE database	Primary customs data cover 12 countries , customs data of the members of ALADI countries, UN COMTRADE database . Container shipping rates from UNCTAD, and various sources are used to compliment the customs data for estimating logistics costs for the rest of the countries	Official national statistics across countries (Census Bureau, and Statistics department). Primary trade statistics data are obtained for 33 countries . The rest of the datasets are estimated.	UN COMTRADE database : Import and export customs specified in HS 6 digit commodity classification with a total of 237 countries which provide varying degree of information regarding CIF, FOB, quantity and mode of transport.	the WITS-Comtrade database, export data for 180 countries

Output variables	Values of import and export of goods in USD for manufacturing and primary products	1) Maritime logistics cost: expressed in USD (CIF-FOB) at 2 and 6 digit HS 1998 classification. There are 97 and 5300 products at 2 and 6 digit HS classification 2) Unit logistics cost: logistics cost per kilogramme, 3) Ad valorem equivalent: logistics cost divided by the import value	Ad valorem costs for trade not broken down by modes : CIF-FOB ratio (CIF/FOB) for 1200 products based on 4- digit HS 2007 classification	1) mode-specific logistics costs, for more than 5000 HS6 digit commodity specification, expressed in USD (CIF-FOB value) 2) Unit logistics costs: logistics cost per kilogramme 3) Mode of transport disaggregated at origin, destination, and HS-6 commodity specification level	Measures of trade costs or comprehensive trade costs for agriculture and manufacturing sectors
Estimation made	none	1) Estimation of logistics costs based on the difference between known CIF and FOB (CIF-FOB) data. Customs data are used for 12 countries. 2) Estimation for transport unit cost per containerized product 3) Ad valorem equivalent costs based on average shipping rates	Estimation of CIF-FOB margin based on a gravity model taking into account socio-economic and geographical variables (distance, oil price, contiguity) for trade between countries	1) Estimation on modal shares, using a multinomial logit model 2) Estimation on transport unit cost, based on a regression model 3) Estimation of maritime logistics costs, using a regression model. The estimation is done based on socio-economic and geographical variables for trade between countries which miss these variables. The estimation model is built on the 2016-2021 trade data	Estimation of measures of trade costs based on a gravity model equation for countries which data are available
Years considered	1995-2018	1991-2007	1995-2020	2016 - 2021	1995-2021

Geographical coverage	40 ALADI (Latin American Integration Association) countries with varying completeness of information provided	43 importing countries from 218 countries of origin at the detailed commodity (6 digit) level of the Harmonized System 1988	190 importing countries and 240 exporting countries, at the detailed commodity (4 digit) level of the Harmonized System 2007	33 importing countries and 237 exporting countries (including ones for which imputation were performed) at the detailed commodity (6 digit) level of the Harmonized System	180 exporting and importing countries
LLDCs covered	2			32	

A few observations can be made in relation to the objective of estimating logistics costs of LLDCs.

1. Among all the datasets, only OECD maritime logistics cost and UNCTAD T&T datasets provide logistics costs data by modes. While OECD dataset only covers maritime logistics costs data until 2007, UNCTAD T&T dataset has more recent data (i.e., until the year 2021) and it has the largest coverage in terms of number of countries, modes, and commodity groups.
2. OECD International Transport and Insurance Costs (ITIC) dataset has a high geographical coverage, but the imputation of the missing costs data seems to have a rather poor performance and a high error margin as reported in (citation).
3. WB-UNESCAP trade costs database includes only 2 commodity groups which limits the granularity of analysis that can be done with the database. Moreover, the calculation of trade costs also depends strictly on the available bilateral trade information among countries included in the dataset. Consequently, trade costs data for LLDCs are also incomplete. That is, only trade costs of LLDCs which report their trade activities with their partners will be available in the database.
4. All databases suffer from varying degrees of incompleteness, especially for countries where customs data are not reported in the UN COMTRADE database. To fill this important gap in data points, typically a model-based imputation is carried out. The two most common models used are gravity and regression models.

The UNCTAD T&T dataset was chosen as the primary data source for this study due to its extensive coverage, more recent data (up to 2021), and detailed granularity in logistics costs, disaggregated by origin, destination, commodity, and mode of transportation. This level of detail allows the dataset to capture the impact of these factors on logistics costs, making it well-suited for the enhanced methodology.

3. Detailed Methodology

3.1 Modelling framework to estimate unit logistics costs broken down by origin, destination, commodity, and mode

The primary aim of this estimation is to develop an Origin-Destination-Commodity-Mode (O-D-C-M) matrix that captures unit logistics costs (measured in \$/ton) for exports and imports between each country and the rest of the global market, categorized by commodity and mode of transport, for the year 2021. While air and sea transport can theoretically connect any pair of countries worldwide, road and rail modes are limited by geographical constraints, particularly the inability to traverse seas. Consequently, some country pairs are associated with four transport modes, while others are restricted to only two. The comprehensive matrix includes unit logistics costs for trade involving 227 countries, across four transport modes and 19 commodity categories, amounting to 2,621,696 data points. These values are determined using FlowMax, which serves as a critical data source for generating the full matrix.

The UNCTAD T&T dataset was selected as the primary data source. However, even though it covers 222 countries as exporters and 169 as importers, the coverage of UNCTAD T&T dataset is insufficient because the recorded country pairs do not cover all commodities and modes. Overall, 72% of the full dataset is missing. Furthermore, the UNCTAD T&T records the transport costs in terms of CIF-FOB margin (in \$) which does not take into account the transport costs and time costs of hinterland transport such as port of arrival to the destination. Therefore, we developed an imputation method to estimate the missing data and include the port-hinterland costs, the framework of which is illustrated in Figure 1. This section describes the methodology along with its strengths and limitations.

As illustrated in Figure 1, the primary data sources for this study are the UNCTAD T&T dataset and FlowMax. The UNCTAD T&T dataset provided detailed information on logistics costs, estimated trade volumes, modes of transport, and commodities for observed bilateral trade. FlowMax supplemented this data with additional information, such as transport distances, trade imbalances (which require the values of a country's exports and imports with the same trading partner), and port-hinterland details. These critical datasets were used to train a Machine Learning (ML) model.

Given that infrastructure upgrades may shorten transport distances, the output variable for the ML model was defined as unit logistics costs per distance (\$/ton-km). This allows flexibility in adjusting for changes in transport distances when calculating unit logistics costs, without needing to rerun the ML model.

FlowMax also provided estimated trade volumes besides the supplemented information for missing trades to impute their unit logistics costs per distance. The complete matrix of unit logistics costs per distance was then used as input for FlowMax to compute overall unit logistics costs. Additionally, FlowMax calculated costs related to the port-hinterland segment, which were combined with maritime logistics costs. The following section outlines this procedure in detail, step by step.

MODELLING FRAMEWORK

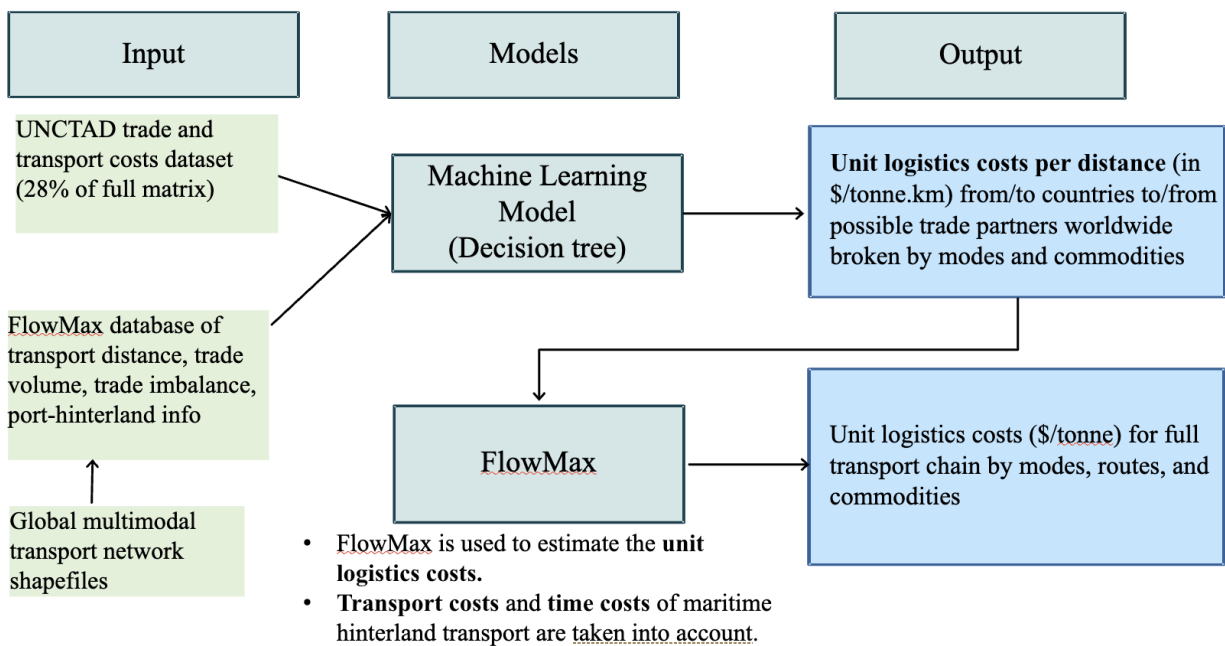


Figure 1 Model Framework

1. Utilize the UNCTAD T&T and FlowMax datasets to build the machine learning model. The primary data source for model building is the UNCTAD T&T dataset spanning 2016 to 2021. This dataset included exporter, importer, commodity, mode, logistics costs (CIF-FOB), and estimated trade volumes.

Since the trade pairs (O-D-C-M) recorded each year vary significantly, incorporating data across all years enhances the input data's heterogeneity. To do this without increasing the model dimension by adding a "Year" variable, we converted data from years other than 2021 to 2021 equivalents using yearly factors.

However, the UNCTAD T&T dataset lacks some crucial information, such as distance, trade imbalance, and port-hinterland distance. To address trade imbalances between OD pairs, we integrated trade estimates from FlowMax. It can also provide distance information, and port-hinterland distances for the observed data. Those data from both UNCTAD T&T dataset and FlowMax were used to train Machine Learning Model to find the correlation between input variables and unit logistics costs per distance (\$/ton-km).

2. Utilize FlowMax database to estimate the missing data

By comparing the UNCTAD T&T dataset with the full matrices, we identified missing data. Given FlowMax's global coverage, we retrieved all input data for these missing bilateral trades. The Decision Tree model was then used to estimate unit logistics costs per distance for these pairs, integrating the retrieved input data. Finally, we combined observed and estimated data to create complete matrices for unit logistics costs per distance.

3. Calculate full chain logistics costs using FlowMax

After estimating the complete unit logistics costs per distance matrices for global trade, we integrated this database back into FlowMax. Using FlowMax's distance data for each OD pair, distinguished by mode and commodity, we measured unit logistics costs (\$/ton), representing the transport expenditure per tonnage. However, the unit logistics costs estimated for sea mode excluded the costs on port-hinterland section, which is a significant part of logistics costs especially for land-locked countries.

We employed the unit logistics costs per distance for countries using road mode (main mode for port-hinterland section) to their neighboring countries as the unit costs for port-hinterland section. This data was used in combination with port-hinterland distance data from FlowMax to measure port-hinterland transport costs for maritime mode. Time costs for the port-hinterland section were also calculated as the product of port-hinterland travel time and the value of time, both sourced from FlowMax. These additional transport and time costs for the port-hinterland section were then added to the maritime logistics costs.

3.2 Decision tree model

A graphical representation of flowchart for a decision tree model is shown in Figure 2, where each internal node signifies the test on an attribute (such as the outcome of a coin flip), each branch denotes the result of the test, and each leaf node signifies a class label or decision made after evaluating all attributes (Sarker, 2021).

For decision tree regression, the structure is similar, but instead of predicting class labels, the leaf nodes represent continuous values or numerical outcomes. For a numerical attribute, the internal node still tests an attribute, but the objective is to split the data in

such a way that minimizes the variance within each partition. The paths from the root to the leaf represent the rules used to make numerical predictions.

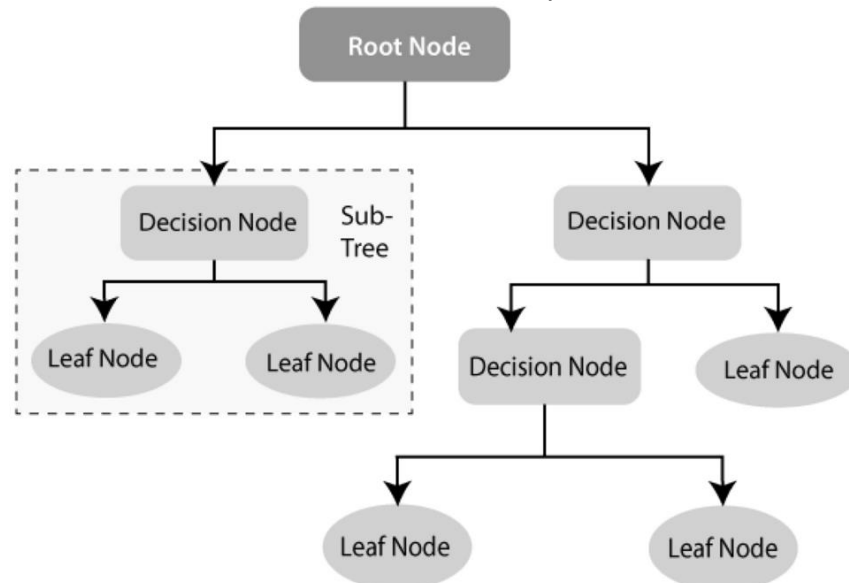


Figure 2 An example of decision tree structure (Sarker, 2021)

To develop a robust prediction model, we considered variables such as bilateral trade volume, travel distances, trade imbalance between import and export of a country (Camisón-Haba & Clemente-Almendros, 2020), transportation mode, commodities, and the product of port-hinterland distance and time and origin/destination country grouping (i.e., 1 means LLDCs and vice versa) to estimate the unit logistics costs per distance (\$/ton*km). These variables are supposed to comprehensively capture the characteristics of global trade that influence transport expenses.

3.3 Strengths and limitations

In this section we explain the strengths and limitations of decision tree model in comparison to the most commonly used methodologies to estimate logistics costs, namely OLS model and gravity model.

3.3.1 Strengths

Decision Tree model is a non-parametric model, capable to capture the non-linear relationship between input and output variables and the interaction among input features, which gives it more flexibility to capture intricate patterns in the data. However, the OLS model assumes a specific parametric form and linear relationship between the independent and dependent variables, which may not always hold true in the logistics costs.

Compared to OLS and machine learning models, the application of gravity model in the WB-UNESCAP database to calculate trade costs is rather indirect. Gravity model is primarily used to predict trade volume, with trade costs as a variable that can be calculated once the model has been estimated. Consequently, the calculation of trade costs in WB-UNESCAP database relies on the availability of trade data for LLDCs to be able to derive the trade costs of LLDCs. Due to the scarcity of trade data for LLDCs, the database does not have a lot of trade costs estimates for these countries and the estimation is limited to two main commodity groups (i.e., agriculture and manufacturing). Another drawback of estimating trade costs as derived output is that it does not use observed logistics costs recorded in customs as input nor output which makes comparison between modelled logistics costs output and observed logistics cost data and to be difficult.

Another strength of our methodology is that the use of distance and time matrix from FlowMax ensures the proper accounting of costs for hinterland transport, which typically form the largest fraction of logistics costs for land-locked countries. This section of the journey is often missing from real-world observation data and logistics cost estimations due to the difficulty in accounting for these costs. The FlowMax, is unique in its capability to distinguish between hinterland and main transport modes, to calculate the additional transport and time costs for the hinterland section next to the estimated logistics costs. This capability allows us to compute the total logistics costs more accurately, including the often-overlooked hinterland section. Furthermore, FlowMax provides granular details of distance and time, distinguished by commodity and mode, enabling the estimation of missing logistics costs by commodity and mode.

3.3.2 Limitations and challenges

A possible limitation of Decision trees is that it can easily overfit the training data, capturing noise and irrelevant details, which can result in poor generalization of new data. By setting the minimum number of samples per leaf and maximum depth of tree, the risk of overfitting can be reduced. It is also hard to interpret the model especially when the tree grows too complex and big, which leads to difficulties in manual estimation, unlike OLS. Also, small changes in the data can result in completely different tree structures, making decision trees unstable and sensitive to variations in the dataset. To overcome this issue, methods such as Random Forests or Gradient Boosting, which aggregate multiple trees are going to be used to enhance stability and accuracy.

Another limitation stems from the incompleteness of UNCTAD T&T dataset. Even though it offers better data coverage than other datasets, 72% of full dataset is still missing. However, this is the nature of empirical trade data, which means all models that use real-world databases typically suffer from the same limitation. Consequently, the quality of the estimation will be likely affected by the limited data coverage.

A challenge in using decision tree model in this study is that the incorporation of the categorical variables, i.e., mode and commodity, as input variables introduces challenges in model accuracy. The incompleteness of UNCTAD T&T dataset makes it even worse. For instance, there are $19 \times 4 = 76$ combinations for mode and commodity such as Food-Road. To avoid overfitting and to ensure the model can capture the potential interaction between commodity and mode, the UNCTAD T&T dataset should have data for each of the C-M combination with a certain frequency. To address this challenge, we implemented the following strategic solution - new data generation:

To address the risk of overfitting caused by missing and low-frequency groupings, we proposed generating plausible new data points that align with the model's assumptions. This involved leveraging data from other years and other groupings and adjusting them according to yearly factors and grouping factors to fabricate data points for combinations that are not observed or in low frequency in the 2021 dataset. This strategy helps fill in the gaps in data continuity and ensures a more comprehensive representation in the model.

3.4 Commodity grouping

The UNCTAD T&T dataset employs the HS-6 commodity classification, resulting in 1,221 recorded commodity groups. However, estimates for the Value of Time, which are necessary for port-hinterland costs calculation, are unavailable for these HS-6 commodities, and the large number of commodity groups increases the complexity of the desired model. To address these challenges, we opted to use 19 commodities from FlowMax, which provides the necessary estimates for the Value of Time. The full list of these commodities and their corresponding HS-6 commodities is provided in Table 2, and the UNCTAD T&T dataset was mapped to align with FlowMax commodities.

Table 2 FlowMax commodities and their corresponding ship types and summary of included HS-6 commodities

ID	Sector/commodities	Ship types	Summary of HS-6 commodities
1	Chemicals and plastic	Wet products	such as industrial chemicals, plastic products, and related chemical materials
2	Refined oil	Oil tanker	such as refined petroleum products, petrochemical derivatives, and other oil-based products utilized in transportation, manufacturing, and consumer goods.
3	Paper wood	dry	such as paper products, wood products, and related materials used in publishing, packaging, and construction.

4	Electronic devices	Container	such as electronic components, consumer electronics, and related equipment
5	Livestock	General cargo	such as various types of livestock, animal products, and related agricultural outputs that are crucial for the food and textile industries
6	Food	Container	such as beverages, tobacco products, and other food-related items such as grape must and products obtained from distilling fermented sugar-cane.
7	Other metals	Dry	such as various non-ferrous metals, metal alloys, and specialized metal products used in industrial applications and manufacturing processes
8	Transport equipment	Vehicle	such as vehicles, transport machinery, and accessories essential for the automotive, aerospace, and logistics industries.
9	Other minerals	Dry	such as various minerals, mineral-based products, and raw materials essential for construction and manufacturing industries
10	Coal	Dry	various types of coal, coal-based products, and other related materials used in energy production and manufacturing.
11	Iron steel	Container	such as iron and steel products, structural components, and related metal materials used in construction and manufacturing.
12	Textile	Container	such as various textile products, finished garments, and textile raw materials used in clothing and upholstery industries.
13	Other agriculture	Dry	such as various types of fish, mollusks, animal products, and aquatic invertebrates. Additionally, it encompasses products like cultured pearls and other miscellaneous agricultural items.
14	Metal products	Container	such as metal tools, fabricated metal products, and machinery components
15	Crude oil	Oil tanker	such as crude petroleum oils, bituminous minerals, and other unrefined petroleum products used as the basis for fuel and industrial processes
16	Other mining	Dry	This category includes a diverse range of items such as precious and semi-precious stones, industrial minerals, and other mined products used in various industries including jewelry, electronics, and construction.
17	Rice crops	Container	such as different types of rice, rice products, and related agricultural commodities critical for global food security.
18	Gas	Gas	such as natural gas, synthetic gas products, and related energy resources utilized for heating, electricity generation, and industrial processes.

19	Other manufacturing	Container	such as tools, domestic appliances, machinery, parts of machines and equipment, ground flying trainers and parts, helicopters and parts, and miscellaneous manufacturing items.
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3.5 Calculation of port-hinterland costs in maritime logistics costs

The application of port-hinterland unit logistics costs is described in detail as follows. A multimodal transport chain typically occurs in maritime transport, which carries out 80% of the international freight transport (UNCTAD, 2021), compared to road, railway, and air. Figure 3 visualizes a schematization of components of a full transport chain for maritime transport. This transport route comprises of hinterland transport using road or rail, transshipment at port, and maritime transport. The costs coming from port-hinterland section are the major reasons why some land-locked countries face bigger challenges than coastal countries, which, however, are always ignored in other studies and are also not recorded in UNCTAD T&T dataset. In this study, port-hinterland unit logistics costs in international maritime trade were calculated by accounting for both time costs and distance-based costs. Detailed formulas for these calculations are provided in section 3.6.

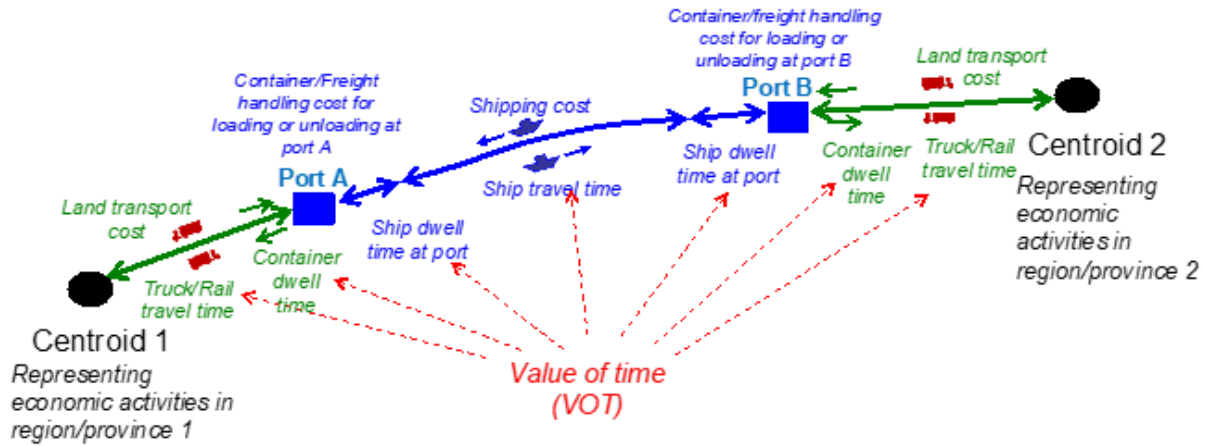


Figure 3 Schematization of components of intermodal logistics costs (source: EMC)

3.6 Calculation of unit logistics costs

The involved formulas to calculate unit logistics costs from unit logistics costs per distance are shown below:

$$ULC_{odkm} = \alpha_{odkm} \sum_{l \in r} D_l \quad (1)$$

$$ULC_{odk_hinterland} = \alpha_{odk_hinterland} \sum_{l \in r_hinterland} D_l + \beta_k \sum_{l \in r_hinterland} t_l \quad (2)$$

Equation (1) applies to every mode:

ULC_{odkm} = unit logistics cost between origin centroid o and destination centroid d for commodity k and mode m (US\$/tonne),

l = links used by the route r ,

D_l = distance during transportation over link l (km),

α_{odkm} = unit logistics costs per kilometer of commodity k and mode m between origin centroid o and destination centroid d over link l (US\$/tonne.km),

Equation (2) only applies to calculate the unit logistics costs of port-hinterland route r for maritime mode:

$ULC_{odk_hinterland}$ = unit logistics cost between origin o and nearest port in the neighboring country d for commodity k (US\$/tonne),

l = links used by the port-hinterland route r ,

D_l = distance during transportation over link l (km),

t_l = time spent during transportation over link l (hour),

$\alpha_{odk_hinterland}$ = unit logistics costs per kilometer of commodity k for port-hinterland mode (mainly road) between origin centroid o and nearest port in the neighboring country d over link l (US\$/tonne.km),

θ_k = value of transport time of commodity k (US\$/hour.tonne).

4. Data sources

In this section, we describe the crucial data that were used to estimate the full matrix of unit logistics costs around the world in FlowMax and their sources.

4.1 Estimates on transport distance and time

In order to carry out imputation for the missing unit logistics costs per distance data, we computed several key input variables such as transport distance, port-hinterland distance and time. The network calculator module of FlowMax was deployed to perform this calculation. The FlowMax network calculator calculated the median transport distance and time from an origin (a city) in one country to a destination (another city) in other country through main routes and by specific mode(s) of transport. The distance and time of sections between origin/destination and port in maritime mode, shown in Figure 3, were also part of the calculated results. These origin and destination points typically represent economic zones where freight traffic is generated and consumed in a country. The latest version of FlowMax includes 227 countries, which makes the calculation of transport time and distance for these countries possible.

The input for this network calculator is a global multimodal transport network model which consolidated and integrated freight transport networks for different transport

modes from various open sources, in Figure 4. The network model is implemented as a GIS (Geographical Information System) shapefile. We retrieved the network for the individual modes from publicly accessible databases which contain a standardised digital schematization of transport networks consisting of nodes and links and their coordinates which can be interpreted by a GIS software. The GIS data used to construct the global networks are obtained from the following sources:

- The road network information integrates two main sources: Global Roads Open Access Data Set (gROADS)² and OpenStreetMap³.
- The rail network is obtained from the Digital Chart of the World (DCW) project,⁴ integrated and updated with the OpenStreetMap data on rail lines and rail stations as intermodal points of connection between road and rail.
- The maritime routes are obtained from the Global Shipping Lane Network data of Oak Ridge National Labs CTA Transportation Network Group.⁵ The network data are with actual travel times for different sea segments. It is possible to connect this network to ports, based on data from the latest World Port Index Database of the National Geospatial-Intelligence Agency.⁶
- The air network is from the openflights.org⁷ which covers more than 10,000 airports, 67663 routes between 3321 airports on 548 airlines spanning the globe.
- Seaports and airports are physically connected to road and rail networks with data on intermodal dwelling times – the time that the cargo spends in the process of being transferred from one mode to another mode of transport. Travel times by type of infrastructure and dwelling times between transport modes are estimated using average speeds based on available information by region.

² <http://sedac.ciesin.columbia.edu/data/set/groads-global-roads-open-access-v1>

³ www.openstreetmap.org

⁴ <https://maps.princeton.edu/catalog/harvard-dcw-rr-line>

⁵ <http://geocommons.com/datasets?id=25>

⁶ <http://msi.nga.mil/NGAPortal/MSI.portal>

⁷ <https://openflights.org/data.html>

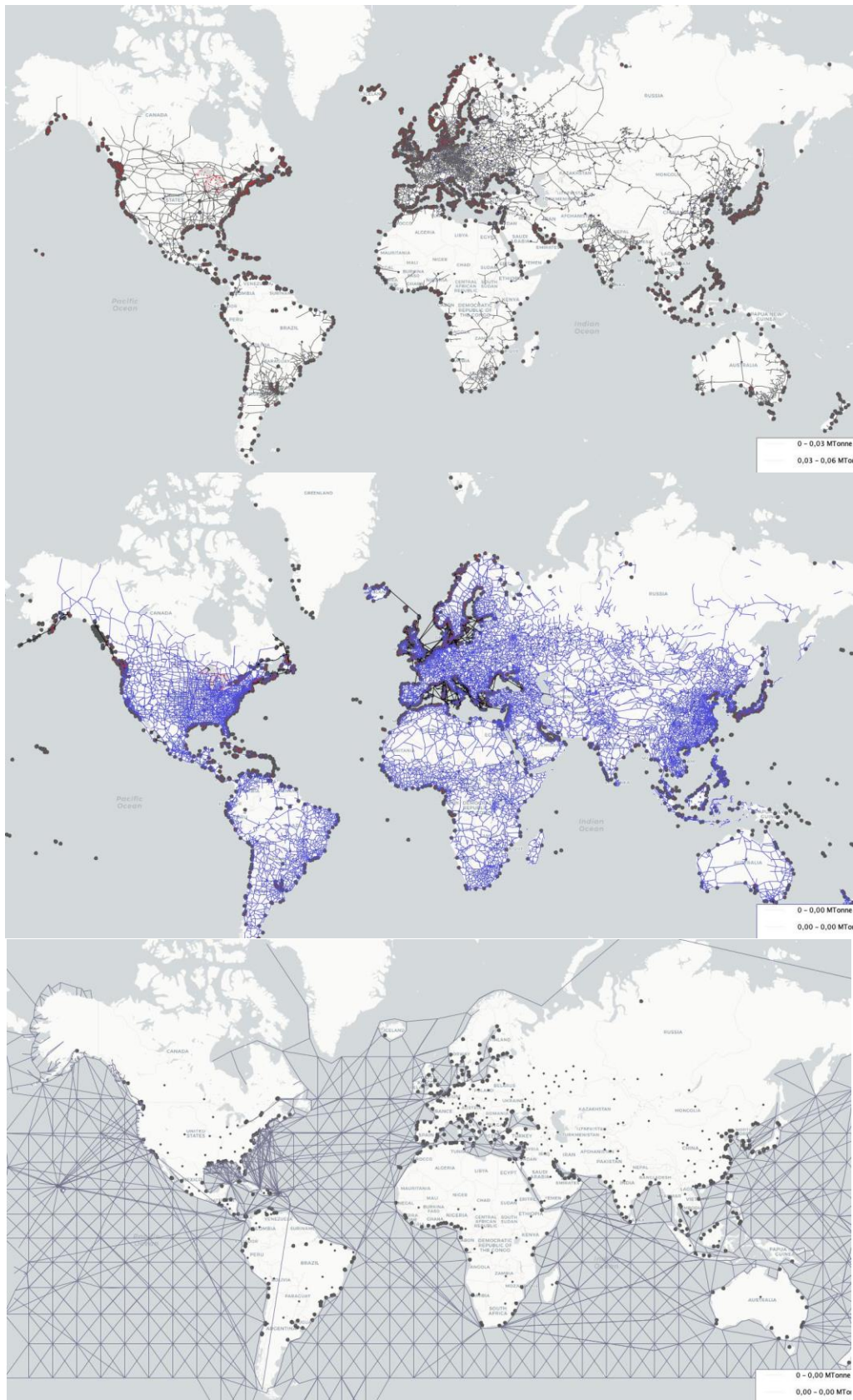




Figure 4 Global freight transport networks (From top to bottom: rail, road, maritime and air networks)

Source: EMC global freight transport networks

Based on the conceptual model above, computation of travel time and travel distance between each origin/destination point for each transport mode (for the modes available for each link) was carried out using shortest path algorithm applied on multimodal transport network model, generating two main outputs:

- The average travel time and distance by mode for each origin destination pair; and
- The shortest path between each origin/destination point for each transport mode.

Similarly, to calculate transport distance and time between ports and hinterland destinations, the model took into account available modes and predict shippers' choice of modes based on distance and time between ports and hinterland destinations. For instance, if intermodal rail transport offers the shortest distance and time, the model will predict that majority of goods will be shipped using rail instead of road.

4.2. Bilateral trade data

Another key input data which was generated by FlowMax is the detailed bilateral trade data in volume by commodity and transport mode, presented at the origin and destination (city) level—a more granular perspective than country-level data. FlowMax could be utilized to predict global trade activities based on the estimates of a macroeconomic trade model such as GTAP (Global Trade Analysis Project). The FlowMax trade database covers 227 countries worldwide (see [Appendix](#), table B 2 for full list of countries). Nevertheless, it is important to note that there is uncertainty in the estimated trade data since there were three steps to obtain the bilateral trade data between origin and destination in volume by commodity and mode:

- 1) disaggregating the output of a macro-economic model which may not represent

sufficient granularity (trade in value by commodity at city level);

- 2) converting disaggregated trade value to trade volume (trade in volume by commodity at city level);
- 3) conducting mode choice by discrete mode choice model (trade in volume by commodity and mode at city level).

By aggregating from the city level to the country level, the results from the first step—disaggregated trade value data—were also used to assess trade imbalances by commodity between country pairs. The final output, comprising bilateral trade data in volume, categorized by mode and commodity, was then utilized as an estimated quantity to support the estimation of missing data.

4.3 UNCTAD Transport and Trade Dataset

The latest version of UNCTAD Trade and Transport (T&T) dataset records international trade flows, in values (in \$) and volumes (estimated, in tonnage), alongside the work and expenditures incurred for their transport (i.e., logistics costs, measured in \$) in Costs of Insurance and Freight (CIF) – Freight On Board (FOB) terms, by pair of trading partners, commodity and mode of transport by which goods entered the destination country, annually, from 2016 to 2021.

5. Results of imputation

5.1 Model performance by commodity

The model's performance is evaluated using the R^2 metric. As displayed in Table 3, the performance varies by commodity, with values ranging from 0.42 to 0.62. This variation reflects differences in the correlation between unit logistics costs per distance and the input variables across different commodities. For example, certain commodities may exhibit lower sensitivity to factors such as trade imbalances or travel distances. Additionally, the availability and quality of empirical data can significantly influence model performance. Some commodities may have fewer observations or lower-quality data, which can result in reduced accuracy for those specific cases.

Table 3 Model performance across commodities

ID	Sector/commodities	Model Performance (R^2)
1	Chemicals_plastic	0.59
2	Refined_oil	0.60
3	Paper_wood	0.61
4	Electronic devices	0.44
5	Livestock	0.42
6	Food	0.42
7	Other_metals	0.54
8	Transport_equipment	0.53

9	Other_minerals	0.55
10	Coal	0.61
11	Iron_steel	0.62
12	Textile	0.49
13	Other_agriculture	0.57
14	Metal_products	0.57
15	Crude_oil	0.62
16	Other_mining	0.42
17	Rice_crops	0.46
18	Gas	0.56
19	Other_manufacturing	0.49

5.2 Global comparison

To demonstrate the utility of the full matrix, we conducted a global comparison to examine variations in unit logistics costs across different country groupings. The country groupings used in this analysis include Landlocked Developing Countries (LLDCs), Other Developing Countries (excluding Least Developed Countries [LDCs], Small Island Developing States [SIDS], LLDCs, and transit countries), Developed Countries (excluding LLDCs), and Transit Developing Countries (including LDCs), for both exports and imports. The comparison is focused exclusively on maritime transportation as this will capture the wide variety of logistics costs across country groupings worldwide the most, including those from LLDCs which typically are disadvantaged due to their high port-hinterland costs.

We present a statistical comparison (in percentage) of unit logistics costs globally in Table 4 where we compare logistics costs of different country groups with LLDCs. Logistics costs of LLDCs are chosen as reference due to their relatively higher costs compared to all country groups. This comparison is based on the median values of maritime unit logistics costs for each country group across reachable trade partner and commodity. Our analysis shows that the differences in unit logistics costs between LLDCs and Transit Countries are the most significant. Our analysis reveals that export costs from LLDCs are 63%, 53% and 44% higher compared to the export costs of Transit Countries, Developed, and Developing economies respectively. On the import side, LLDCs see 75%, 42% and 34% higher import costs compared to Transit countries, Developed, and Developing economies respectively.

Higher export and import costs of LLDCs than other country groups quantitatively present the challenges faced by LLDCs, such as higher production costs, lower competitiveness, reduced integration, and hampered economic development and food security. Higher production costs arise because LLDCs often incur additional transportation and logistical expenses due to their lack of direct access to seaports. This increases the overall cost of

goods, making them less competitive on the international market. Reduced competitiveness further limits their ability to attract foreign direct investment, which is crucial for economic growth and development.

Moreover, the difficulty in integrating into global value chains (GVCs) means that LLDCs face difficulty in seeing the benefits of global trade, such as technology transfer, skills development, and access to larger markets. This hampers their economic diversification and industrialization efforts.

Table 4 Percentage difference of LLDC's unit logistics cost compared to other economic groups

	LLDCs vs Transit	LLDCs vs Developed economies	LLDCs vs Developing economies
Export	63%	53%	44%
Import	75%	42%	34%

5.2.1 Logistics costs of exports

Figure 5 presents the distribution of unit logistics costs of exports for each country group across market and commodity. The lower bound, middle line, and upper bound of the box are quite meaningful. Specifically:

- The lower bound (bottom of the box) is the 25th percentile, meaning 25% of the data is below this value.
- The middle line (inside the box) is the median, meaning 50% of the data is below this value.
- The upper bound (top of the box) is the 75th percentile, meaning 75% of the data is below this value.

LLDCs experience the greatest variation in unit logistics costs compared to other country groups, indicating that the trade challenges they face, the quality of their hinterland transport and their connectivity to the gateway ports vary widely. LLDCs generally incur the highest logistics costs compared to other country groups worldwide. This is demonstrated in both the boxplot visualization and the statistical comparison in Table 4, where LLDCs exhibit the highest median unit logistics costs. Additionally, the lower and upper quartiles of LLDCs' logistics costs are higher than those of other country groups. The logistics costs for developed and developing countries are relatively similar, although developing countries show greater variability. The subsequent analysis reveals their differences in unit logistics costs by commodity. Interestingly, transit countries have even lower logistics costs than developed countries. This could be attributed to the inclusion of many small islands within the developed countries group in this analysis. The data

quality for these islands is often low, and their maritime logistics costs are often high, which increases the median value of logistics costs for developed countries.

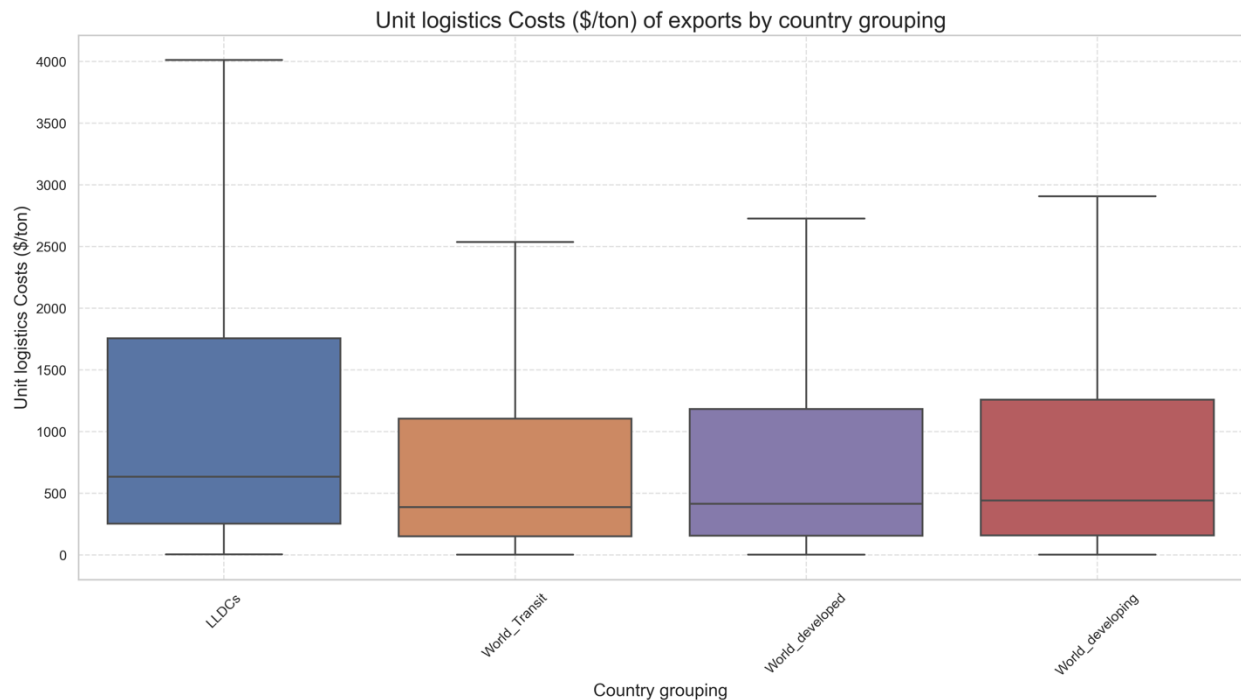


Figure 5 Global comparison on unit logistics costs of exports in the year 2021

Figure 6 presents a global comparison of unit logistics costs for exports by commodity across different markets. LLDCs consistently face the highest unit logistics costs for nearly all commodities. The logistics costs for the top export commodities for LLDCs, namely crude oil, other metals, and food, are unfortunately higher than the other country groupings with an exception to crude oil which is comparable to developed economy. This has indicated the underlying factor which affects export competitiveness of LLDCs compared to other regions globally. It becomes difficult for LLDCs to compete with regions that have better-developed infrastructure and lower logistics costs. This scenario could also deter foreign investment, as investors seek more favorable conditions elsewhere, further limiting economic growth and development.

The most notable cost discrepancies are seen in electronic devices and other manufacturing, with LLDCs' costs being approximately \$2,000 per ton higher than those of transit countries. While transit countries generally have the lowest costs, exceptions are observed in processed commodities such as metal products, transport equipment, chemical plastics, refined oil, iron and steel, and paper and wood, where developed countries incur the lowest costs. Crude oil and coal have the lowest unit logistics costs, likely due to their low logistics costs per distance and minimal time-related expenses.

Overall, developing and developed countries exhibit similar unit logistics costs for most commodities, with notable variations in electronic devices, crude oil, chemical plastics, gas, and paper and wood. The largest cost difference for high-value commodities is found in electronic devices, where developed countries incur higher costs. Conversely, for low-value commodities, the most significant difference is observed in paper and wood, with developing countries facing higher costs. Besides the concerns mentioned above about the classification of developed countries, the higher logistics costs for electronic devices in developed countries can also be attributed to the need for advanced handling and security measures, despite the use of sophisticated logistics technologies, automation, and higher export volumes that typically reduce costs.

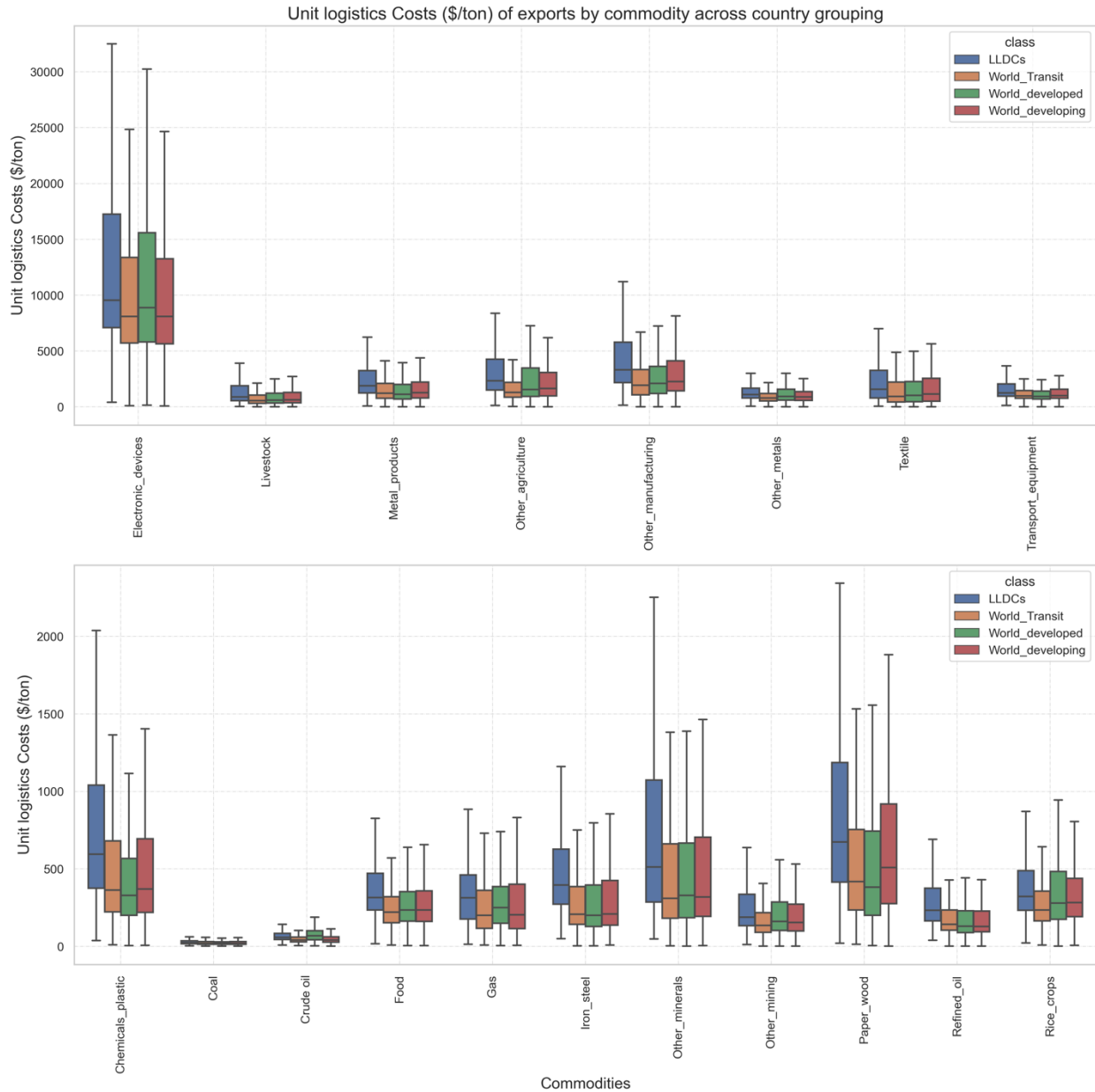


Figure 6 Global comparison on unit logistics costs of exports by commodity in the year 2021

5.2.2 Logistics costs of imports

Figure 7 illustrates the distribution of unit logistics costs of imports for each country group across various markets and commodities. Generally, LLDCs continue to exhibit the highest unit logistics costs for imports. However, the import costs for LLDCs are generally lower than their export costs. Additionally, the costs of imports for transit countries are generally higher than their export costs. Meanwhile, the import costs for developed and developing countries are, on average, similar.

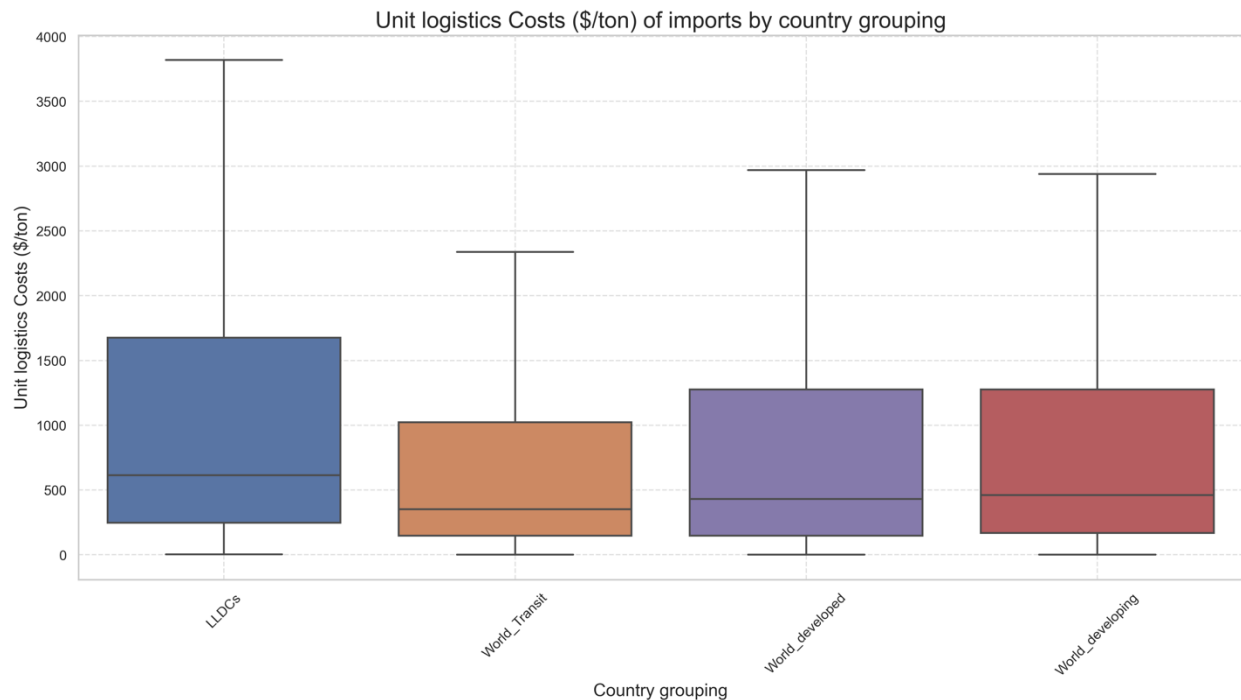


Figure 7 Global comparison on unit logistics costs of imports in the year 2021

Figure 8 presents a global comparison of unit logistics costs for imports by commodity. LLDCs incur the highest costs for the majority of commodities, with the exception of electronic devices and textiles. Notably, LLDCs generally experience lower logistics costs for imports than for exports, with significant differences observed particularly for electronic devices, chemicals and plastics, other metals, and paper and wood. Transit countries typically experience the lowest import costs across most commodities.

For LLDCs, significantly higher import costs for the top three import commodities—other manufacturing, chemicals and plastics, and refined oil—pose significant challenges to economic development, societal well-being, and food security. Elevated costs for other manufacturing goods increase production expenses for local industries, diminishing their competitiveness and raising prices for finished products. Similarly, high costs for importing chemicals and plastics affect various sectors, from agriculture to pharmaceuticals, leading to higher consumer goods prices and potential inflation. The agricultural sector, in particular, relies heavily on chemicals and plastics for fertilizers, pesticides, and packaging, making high import costs a direct threat to food production and availability.

Additionally, high import costs for refined oil can result in increased energy and transportation expenses, impacting nearly every sector of the economy and raising the cost of living. Increased transportation costs can lead to higher food prices, as the cost of transporting agricultural products to markets rises. Collectively, these high import costs

reduce overall economic competitiveness, deter foreign investment, and strain government resources as they attempt to alleviate these impacts through subsidies or policy interventions. Ultimately, this hampers sustainable economic growth and development, and exacerbates food insecurity by increasing the cost and reducing the availability of essential food items.

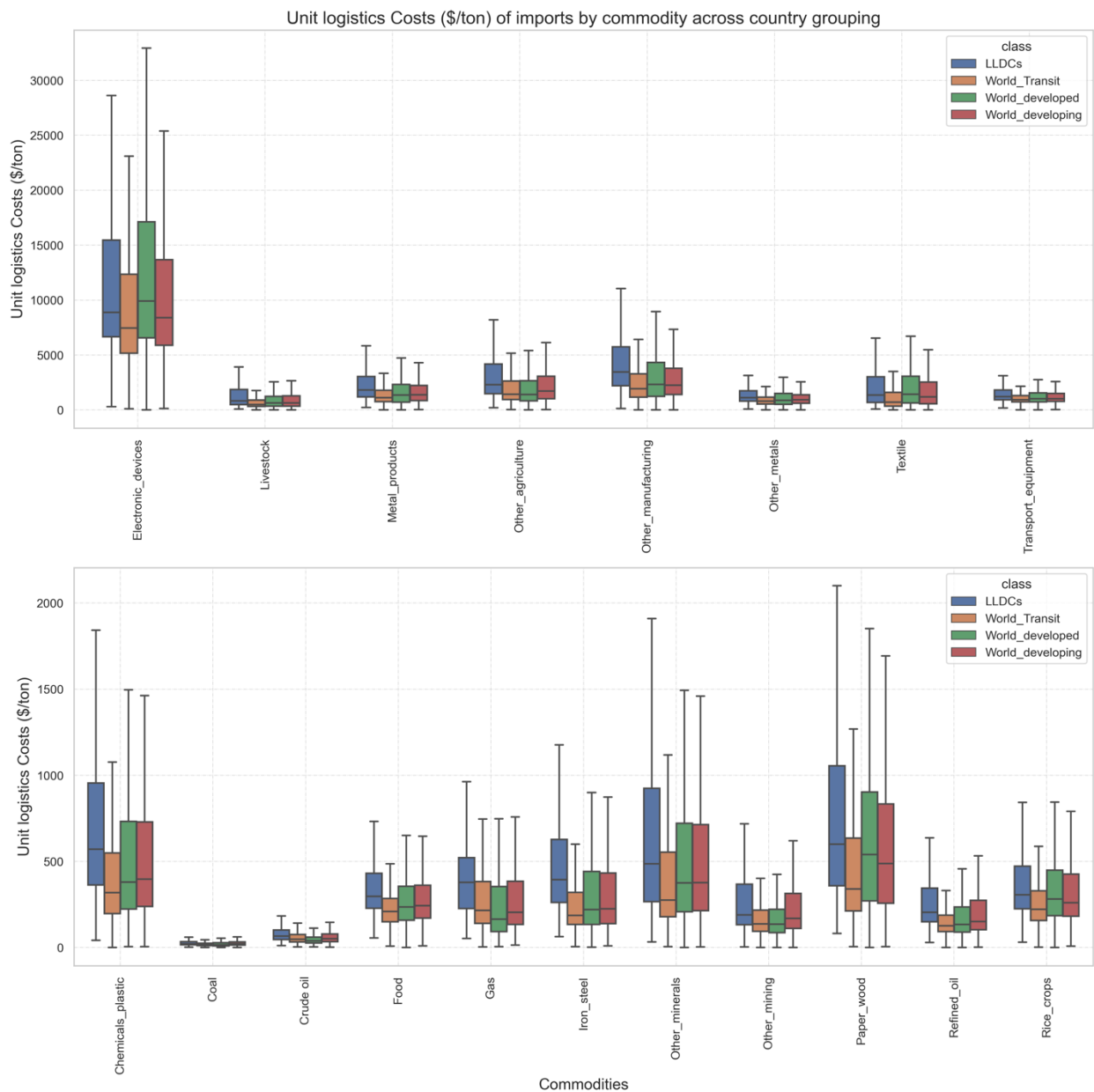


Figure 8 Global comparison on unit logistics costs of imports by commodity in the year 2021

6. Description of FlowMax

FlowMax is an international freight transport model specifically designed to project international freight transport activities (in tonne-kilometres), logistics costs and modal share until 2050 under different trade and policy scenarios. EMC hosts one of the leading international freight transport models that has been validated, published in a scientific journal and applied to project transport activities in different policy analysis projects, including comprehensive impact assessment of IMO measures. The model is highly granular with spatial resolution that covers 227 countries, 431 regions, and 5000 ports worldwide. It is built on the four-steps freight transportation modelling approach, and it takes the global trade projection from a macroeconomic model such as GTAP as an input. Figure 9 provides a conceptual design of the framework. FlowMax is designed to be able to estimate the weight of commodities traded between countries, the choice between modes and transport routes used to transport these commodities based on transport networks characteristics, and relevant socio-economic variables such as logistics costs and time. The model consists of the following components:

- **Trade Disaggregation model** which disaggregates international trade values from economic trade model into into a larger number of production/consumption centroids. These centroids were calculated for all the cities around the world classified by United Nations in 2010 relative to their population (2539 cities). This resulted in 431 centroids globally, with spatially balanced results also for all continents.
- **Value-to-Weight model** which estimates the rate of conversion of value units (dollars) into weight units of cargo (tonnes) by mode, calibrated using datasets from Eurostat and Economic Commission for Latin America and the Caribbean (ECLAC) data on value/weight ratios for different commodities.
- **Mode Choice model** which predicts the modal share of the transport modes used for trade between any origin–destination pair of centroids. The model includes 5 major modes: air, rail, road, sea, and waterways. The mode attributed to each trade connection represents the longest transport section. All freight will require intermodal transport both at the origin and destination. This domestic component of international freight is usually not accounted for in the literature but is included in our model to account for realistic costs for firms. The model is estimated using a standard multinomial logit estimator including commodity type panel terms on travel times and cost.
- **Route Choice model** which predicts the share of shipping volume assigned across all possible international shipping routes between all origins and destinations. We use a path size logit model in combination with a path generation method to assign the volume of freight transport. The model does this using a shortest path

algorithm and choice set creation algorithm to identify the subsegments of the complete shortest route for each port-to-port segment of a shipping line. Using this model annual transport work for different ship types and commodities are computed for 19 commodities.

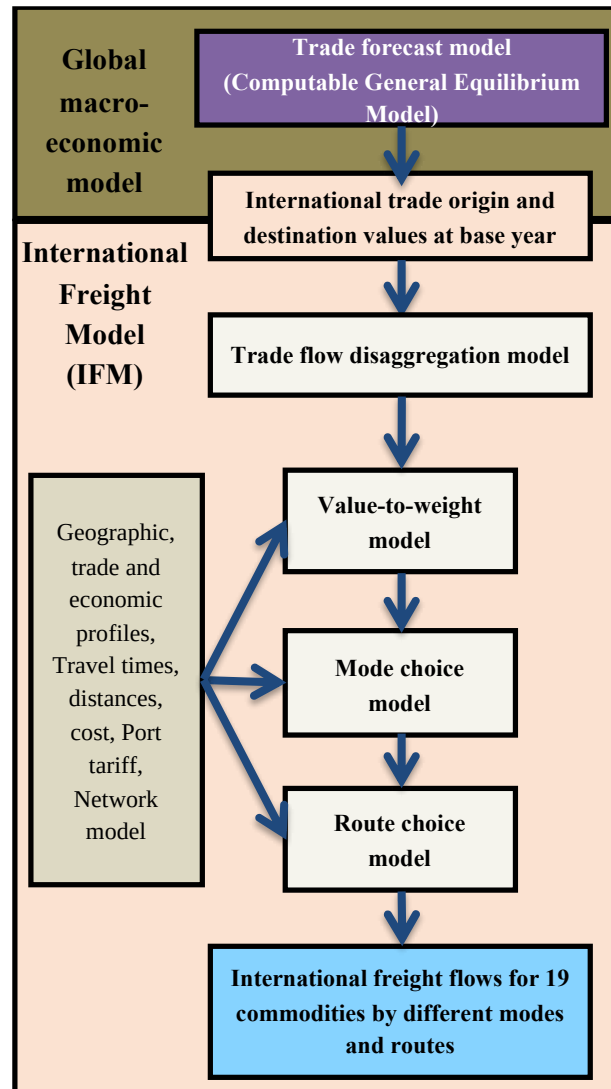


Figure 9 International Freight Model (Source: Halim et al., 2018)

Appendix

Table B 1 Model performance across commodities

ID	Sector/commodities	Model Performance
1	Chemicals_plastic	0.59
2	Refined_oil	0.60
3	Paper_wood	0.61
4	Electronic devices	0.44
5	Livestock	0.42
6	Food	0.42

7	Other_metals	0.54
8	Transport_equipment	0.53
9	Other_minerals	0.55
10	Coal	0.61
11	Iron_steel	0.62
12	Textile	0.49
13	Other_agriculture	0.57
14	Metal products	0.57
15	Crude oil	0.62
16	Other_mining	0.42
17	Rice_crops	0.46
18	Gas	0.56
19	Other_manufacturing	0.49

Table B 2 FlowMax country list

ID	Country list	ISO list
1	Japan	JPN
2	Malaysia	MYS
3	Taiwan	TWN
4	Tajikistan	TJK
5	Ukraine	UKR
6	Mexico	MEX
7	Moldova	MDA
8	Nicaragua	NIC
9	Nigeria	NGA
10	Pakistan	PAK
11	Panama	PAN
12	Paraguay	PRY
13	Peru	PER
14	Philippines	PHL
15	Australia	AUS
16	Austria	AUT
17	Azerbaijan	AZE
18	Bangladesh	BGD
19	Belgium	BEL
20	Bolivia	BOL
21	Brazil	BRA
22	Bulgaria	BGR
23	Cambodia	KHM
24	Cameroon	CMR

25	Canada	CAN
26	China	CHN
27	Czech Republic	CZE
28	Denmark	DNK
29	Dominican Republic	DOM
30	Ecuador	ECU
31	Egypt	EGY
32	Ethiopia	ETH
33	Finland	FIN
34	France	FRA
35	Guinea	GIN
36	Haiti	HTI
37	Honduras	HND
38	China_Hong Kong	HKG
39	Hungary	HUN
40	India	IND
41	Belarus	BLR
42	Colombia	COL
43	Germany	DEU
44	Indonesia	IDN
45	Iran	IRN
46	Italy	ITA
47	Mongolia	MNG
48	Morocco	MAR
49	Mozambique	MOZ
50	Myanmar	MMR
51	Netherlands	NLD
52	New Zealand	NZL
53	Afghanistan	AFG
54	Algeria	DZA
55	AngXSM	AGO
56	Argentina	ARG
57	Armenia	ARM
58	Chad	TCD
59	Chile	CHL
60	Cote d'Ivoire	CIV
61	Croatia	HRV
62	Cuba	CUB
63	Georgia	GEO

64	Ghana	GHA
65	Greece	GRC
66	Guatemala	GTM
67	Jamaica	JAM
68	Jordan	JOR
69	Kazakhstan	KAZ
70	Kenya	KEN
71	Russia	RUS
72	Saudi Arabia	SAU
73	Senegal	SEN
74	Spain	ESP
75	Sri Lanka	LKA
76	Sudan	SDN
77	Sweden	SWE
78	Syria	SYR
79	Tanzania	TZA
80	United Kingdom	GBR
81	United States	USA
82	Uzbekistan	UZB
83	Venezuela	VEN
84	Vietnam	VNM
85	Yemen	YEM
86	Zambia	ZMB
87	Zimbabwe	ZWE
88	Macedonia	MKD
89	Montenegro	MNE
90	Slovakia	SVK
91	Slovenia	SVN
92	Turkmenistan	TKM
93	Madagascar	MDG
94	Iceland	ISL
95	United Arab Emirates	ARE
96	Iraq	IRQ
97	Ireland	IRL
98	Latvia	LVA
99	Lebanon	LBN
100	Lithuania	LTU
101	Mali	MLI
102	PXSMnd	POL

103	Portugal	PRT
104	Congo	COG
105	Romania	ROU
106	Serbia	SRB
107	Singapore	SGP
108	Somalia	SOM
109	South Africa	ZAF
110	South Korea	KOR
111	Thailand	THA
112	Tunisia	TUN
113	Turkey	TUR
114	Uganda	UGA
115	Uruguay	URY
116	Norway	NOR
117	Albania	ALB
118	Bosnia & Herzegovina	BIH
119	Switzerland	CHE
120	Cyprus	CYP
121	Estonia	EST
122	Kyrgyzstan	KGZ
123	Malta	MLT
124	Brunei	BRN
125	Central African Republic	CAF
126	Congo, the Democratic Republic of the	COD
127	Papua New Guinea	PNG
128	Namibia	NAM
129	Laos	LAO
130	Puerto Rico	PRI
131	Qatar	QAT
132	Solomon Island	SLB
133	Bahrain	BHR
134	Malawi	MWI
135	Mauritius	MUS
136	Nepal	NPL
137	Togo	TGO
138	Trinidad and Tobago	TTO
139	Benin	BEN
140	Anguilla	AIA
141	Aruba	ABW

142	British Virgin Islands	VGB
143	Barbados	BRB
144	Saint Kitts and Nevis	KNA
145	Saint Lucia	LCA
146	Montserrat	MSR
147	Turks and Caicos Islands	TCA
148	Saint Vincent and the Grenadines	VCT
149	Virgin Islands	VIR
150	Eritrea	ERI
151	Sao Tome and Principe	STP
152	Bahamas	BHS
153	Belize	BLZ
154	Bermuda	BMU
155	Bhutan	BTN
156	Burundi	BDI
157	Comoros	COM
158	Djibouti	DJI
159	French Polynesia	PYF
160	Gambia	GMB
161	Grenada	GRD
162	Seychelles	SYC
163	Cayman Islands	CYM
164	Fiji	FJI
165	Kiribati	KIR
166	Marshall Islands	MHL
167	Nauru	NRU
168	New Caledonia	NCL
169	Niue	NIU
170	Palau	PLW
171	Samoa	WSM
172	Tonga	TON
173	Tuvalu	TUV
174	Vanuatu	VUT
175	American Samoa	ASM
176	Antigua and Barbuda	ATG
177	Maldives	MDV
178	Lesotho	LSO
179	Guadeloupe	GLP
180	Timor-Leste	TLS

181	Suriname	SUR
182	Cabo Verde	CPV
183	Commonwealth of Northern Marianas	MNP
184	Cook Islands	COK
185	Dominica	DMA
186	Federated States of Micronesia	FSM
187	Guinea-Bissau	GNB
188	Liberia	LBR
189	Mauritania	MRT
190	Niger	NER
191	Sierra Leone	SLE
192	Guyana	GUY
193	Luxembourg	LUX
194	Burkina Faso	BFA
195	Rwanda	RWA
196	Andorra	AND
197	Bonaire	BES
198	Botswana	BWA
199	China_Macao SAR	MAC
200	Christmas Island	CXR
201	Costa Rica	CRI
202	Curacao	CUW
203	El Salvador	SLV
204	Equatorial Guinea	GNQ
205	Eswatini	SWZ
206	Falkland Islands (Malvinas)	FLK
207	Faroe Islands	FRO
208	Gabon	GAB
209	Gibraltar	GIB
210	Greenland	GRL
211	Guam	GUM
212	Israel	ISR
213	Korea_Dem. People's Rep	PRK
214	Kuwait	KWT
215	Libya	LBY
216	Norfolk Island	NFK
217	Oman	OMN
218	Saint Barthelemy	BLM
219	Saint Helena	SHN

220	Saint Martin (French part)	MAF
221	Saint Pierre and Miquelon	SPM
222	San Marino	SMR
223	Sint Maarten (Dutch part)	SXM
224	South Sudan	SSD
225	State of Palestine	PSE
226	Tokelau	TKL
227	Western Sahara	ESH